

MSSC (Microbial Sump-to-Soil Cultivator)

Prior Art & IP Defense Dossier

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1. Executive Summary

- Earliest infrastructure groundwork: duck pond + bog filter + multi-species manures discussed March 22, 2025.
- First explicit MSSC framing: March 23–24, 2025 timeframe.
- Compost tea & microbial cycling detailed through late March, including ducks as biological primers (March 28, 2025).
- Patent-style draft authored June 8–10, 2025; experimental logs initiated June 10, 2025.

2. Chronological Timeline & Evidentiary Index

Date	Record / Evidence Summary
2025-03-22	Duck pond + bog filter under construction; access to multi-species manures (duck, chicken, go
2025-03-23	MSSC framing begins (sump–soil cycling, compost tea intention).
2025-03-24	MSSC explicitly acknowledged as coming into fruition.
2025-03-28	Ducks as biological primers for rows—rotation-of-life concept.
2025-06-08	Rewrite MSSC wiki w/ validated components; align with Tri-Source.
2025-06-09	HomeNode concept: residential nitrate removal using MSSC biome.
2025-06-10	Lab tests: small containers; begin documentation & scale plan.

3. Patent-Style Specification (June 2025)

Field

Agricultural microbiology; integrated water treatment and soil amendment systems.

Summary

MSSC bioreacts runoff via sump/bog reactor and returns inoculated irrigation; captures compost tea stream.

Representative Embodiment

60-ft pond, 10×40-ft bog, aeration, dosing ports, recirc pump, drip tee, compost tea bleed.

Controls

DO/ORP/EC/pH/temp sensors; stage-specific logic.

Duck-Row Priming

Rotational duck priming to establish beneficial consortia.

Exhibit A — Overlap Matrix: Carbon Mining Exchange vs. MSSC Prior Art

Accessed: August 27, 2025 (America/Chicago)

Website Claim (verbatim excerpt)	URL/Section	Observed	Overlap with MSSC	Relevant MSSC Date(s)	Notes
...r microbial sump system alone has proven significantly more effective than traditional aerobic systems in reducing methane emissions from agricultural waste.	Homepage	2025-06-27	Higher potential	2025-03-28 (MSSC)	Direct overlap
...mbiotic Sump Technology — A proprietary system that utilizes a diverse community of microorganisms to break down organic matter, producing a nutrient-rich sludge and a clear, odorless liquid.	Technology page	2025-08-27	Higher-oxygenated	2025-03-23, 2025-06-08	Proprietary
...ansforms the farm pond into an ecological asset, generating revenue through the sale of carbon credits and the production of high-quality organic fertilizers.	Technology page	2025-08-27	Medium-to-High	2025-03-22 (MSSC)	Proprietary
...arbon Coherence Score" used to determine the environmental impact of various agricultural practices, providing farmers with actionable insights to optimize their operations.	Technology page	2025-08-27	Medium	2025-03-22 (MSSC)	Proprietary
...ys farmers up to \$10,000 per acre... verified carbon credits.	Home page	2025-08-27	Low-to-Medium	2025-06-08 onward	Conceptual overlap
...land-grab and speculation in the carbon market.	Home page	2025-08-27	Low-to-Medium	2025-06-08 onward	Conceptual overlap

Conclusion: Multiple statements on the Carbon Mining Exchange website overlap materially with preexisting MSSC concepts, terminology, and system architecture documented in our logs and June 2025 patent-style draft.